



Semester 2 Examinations 2013/ 2014

Exam Code(s) 3BME1, 3BMS2, 3FM2, 3MR3, 3BS3, 3BS9,
4BS4, 4BMS2

Exam(s) 3rd Arts, 3rd & 4th Science

Module Code(s) MP307
Module(s) Modelling II

Paper No. 1
Repeat Paper No

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Internal Examiner(s) Professor M. Destrade
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Instructions: For full marks answer question 1 in Section A **and**
TWO questions in Section B correctly.
All questions carry the same weight.

Duration 2hrs
No. of Pages 6
Department(s) Applied Mathematics
Course Co-ordinator(s) Dr P.T. Piironen

Requirements:

Calculator Yes
Statistical/ Log Tables Yes
Release to Library Yes

Section A - Answer all questions (a)-(d). (Each question is worth **5 Marks**.)

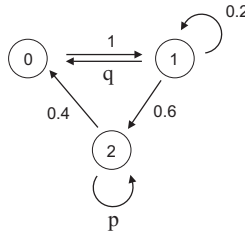
1. (a) Consider a queue and let

$$f(t) = \begin{cases} \frac{1}{b-a}, & a \leq t \leq b, \\ 0, & t < a, t > b, \end{cases} \quad (\text{Uniform distribution})$$

be the probability that a customer has to wait t minutes before being served. Assume a and b are positive constants and $a < b$.

- (i) What is the average waiting time?
- (ii) What is the variance and the standard deviation of the waiting time?

- (b) Consider the following Markov Process with transition probabilities shown.



- (i) Find the transition matrix P and the probabilities p and q .
- (ii) Is the Markov process ergodic? Explain your answer.
- (iii) Find the equilibrium probabilities.

- (c) Consider the nonlinear population model

$$P_{n+1} = R(P_n)P_n,$$

where P_n is the population at time n and

$$R(P) = \frac{1+r}{1+r\frac{P}{K}}, \quad r, K > 0.$$

Find the fixed points for this model and show for what initial populations $P_0 \geq 0$ the population will survive and disappear, respectively.

(d) Consider the linear control system

$$\begin{aligned}\frac{dx_1}{dt} &= b_1 x_1, \\ \frac{dx_2}{dt} &= x_1 + x_2 + u, \\ y &= x_1,\end{aligned}$$

where $x_1 \in \mathbb{R}$ and $x_2 \in \mathbb{R}$ represent the state, $u \in \mathbb{R}$ is the input, $y \in \mathbb{R}$ is the output and $b_1 \in \mathbb{R}$ is the control parameter.

(i) Is this system observable? Explain your answer.

(ii) Is this system reachable? Explain your answer.

Section B - Answer any two questions.

2. Consider a **two-state** queue (eg. a single telephone line) with arrival and service patterns described by Poisson processes with parameters α and β , respectively. Let

$$p_k(t) = \text{Prob}(k \text{ customers queuing at time } t), \quad k = 0, 1.$$

- (a) Sketch a diagram showing all possible transitions.

2 Marks

- (b) Show that the probabilities $p_0(t)$ and $p_1(t)$ evolve according to the differential equations

$$\begin{aligned} \frac{dp_0(t)}{dt} &= -\alpha p_0(t) + \beta p_1(t), \\ \frac{dp_1(t)}{dt} &= \alpha p_0(t) - \beta p_1(t). \end{aligned}$$

8 Marks

- (c) Calculate the equilibrium probabilities π_0 and π_1 for the two-state queue.

4 Marks

- (d) Consider a single telephone line and let 6 minutes be the average time for a call to arrive and let the average length of a call be 2 minutes. What is the probability that the phone line is busy?

6 Marks

3. Consider an ecological system with a prey population of size $x(t)$ and a predator population of size $y(t)$ which satisfy the differential equations

$$\begin{aligned}\frac{dx}{dt} &= x(2 - y), \\ \frac{dy}{dt} &= y(-3 + x).\end{aligned}$$

- (a) Give an interpretation for the various terms in these equations.

4 Marks

- (b) Show that the equilibrium points are $(x, y) = (0, 0)$ and $(x, y) = (3, 2)$. For small deviations away from the equilibrium point $(x, y) = (3, 2)$, show that the system is periodic with period $\sqrt{6}\pi/3$.

8 Marks

- (c) Show that in general the solution satisfies

$$K = \frac{e^{x+y}}{x^3 y^2}$$

for some constant K . Sketch a plot of x vs. y and comment on the typical behaviour of this system in an application of this model.

8 Marks

4. A two-compartment model for drug delivery can be modelled as

$$\begin{pmatrix} \dot{c}_1 \\ \dot{c}_2 \end{pmatrix} = \begin{pmatrix} -p_0 - p_1 & p_1 \\ p_2 & -p_2 \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} + \begin{pmatrix} b_0 \\ 0 \end{pmatrix} u, \quad y = c_2,$$

where c_1 and c_2 are the concentrations of the drug in each compartment, p_0 , p_1 , p_2 and b_0 are rate constants, u is the flow rate of the drug into compartment 1 and y is the sum of the concentrations of the drug in compartments 1 and 2.

- (a) Draw a schematic diagram of this system including concentrations, flow directions and rate parameters.

4 Marks

- (b) Investigate for what values of p_0 , p_1 and p_2 this system is observable.

6 Marks

- (c) Design a control function

$$u = -Kc + k_r r, \quad \text{for } K = \begin{pmatrix} k_1 & k_2 \end{pmatrix}, \quad c = \begin{pmatrix} c_1 \\ c_2 \end{pmatrix},$$

where k_1, k_2, k_r are the control parameters, such that the desired output $r = 4$ and the closed loop eigenvalues for the system satisfies $\lambda_1 = -3$ and $\lambda_2 = -1$.

10 Marks